6-1. Multiple regression: Parameter estimation

Given response variables Y_i $(i = 1, 2, \ldots, n)$ and explanatory variables/predictors $X_1, X_2, \ldots, X_k$, the linear regression model is

$$Y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_k x_{ik} + \epsilon_i \qquad (i = 1, 2, \ldots, n)$$

with $E[\epsilon_i] = 0$ and $\text{Var}[\epsilon_i] = \sigma^2$ for $i = 1, \ldots, n$.

There are $k + 2$ parameters: $\beta_0, \beta_1, \ldots, \beta_k$ and σ^2

Note that β_0 is the intercept term, formerly denoted a

In matrix notation, the model can be written as

$$\boldsymbol{Y} \;=\; X\boldsymbol{\beta} + \boldsymbol{\epsilon}$$

where

$$\boldsymbol{Y} = \begin{pmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_n \end{pmatrix}, \qquad X = \begin{pmatrix} 1 & x_{11} & x_{12} & \cdots & x_{1k} \\ 1 & x_{21} & x_{22} & \cdots & x_{2k} \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & x_{n1} & x_{n2} & \cdots & x_{nk} \end{pmatrix},$$

$$\boldsymbol{\beta} = \begin{pmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_k \end{pmatrix}, \qquad \boldsymbol{\epsilon} = \begin{pmatrix} \epsilon_1 \\ \epsilon_2 \\ \vdots \\ \epsilon_n \end{pmatrix}.$$

X is called the **design matrix**.

Note: intercept is embedded in the vector $\boldsymbol{\beta}$

To estimate $\beta_0, \ldots, \beta_k$ we use OLS (Ordinary Least Squares).
Minimise square of residuals S:

$$\begin{aligned} S &= (\boldsymbol{Y} - X\boldsymbol{\beta})^T (\boldsymbol{Y} - X\boldsymbol{\beta}) \\ &= \left(\boldsymbol{Y}^T - \boldsymbol{\beta}^T X^T\right) (\boldsymbol{Y} - X\boldsymbol{\beta}) \\ &= \boldsymbol{Y}^T\boldsymbol{Y} - \boldsymbol{\beta}^T X^T\boldsymbol{Y} - \boldsymbol{Y}^T X\boldsymbol{\beta} + \boldsymbol{\beta}^T X^T X\boldsymbol{\beta} \\ &= \boldsymbol{Y}^T\boldsymbol{Y} - 2\boldsymbol{\beta}^T X^T\boldsymbol{Y} + \boldsymbol{\beta}^T X^T X\boldsymbol{\beta} \\ \Rightarrow \quad \frac{\partial S}{\partial \boldsymbol{\beta}} &= -2X^T\boldsymbol{Y} + 2X^T X\boldsymbol{\beta} \end{aligned}$$

At a stationary point,

$$\begin{aligned} \frac{\partial S}{\partial \boldsymbol{\beta}} &= \mathbf{0} \\ \Rightarrow \quad 2X^T\mathbf{Y} &= 2X^TX\hat{\boldsymbol{\beta}} \\ \Rightarrow \quad & \boxed{\hat{\boldsymbol{\beta}} = \left(X^TX\right)^{-1}X^T\mathbf{Y}} \end{aligned}$$

provided X^TX is non-singular.

Check for bias:

$$\begin{aligned} \hat{\boldsymbol{\beta}} &= \left(X^T X\right)^{-1} X^T \boldsymbol{Y} \\ &= \left(X^T X\right)^{-1} X^T \left(X\boldsymbol{\beta} + \boldsymbol{\epsilon}\right) \\ &= \boldsymbol{\beta} + \left(X^T X\right)^{-1} X^T \boldsymbol{\epsilon} \\ \Rightarrow \quad E\left[\hat{\boldsymbol{\beta}}\right] &= \boldsymbol{\beta} + \left(X^T X\right)^{-1} X^T \, E\left[\boldsymbol{\epsilon}\right] \\ &= \boldsymbol{\beta} \end{aligned}$$

That is, $\hat{\boldsymbol{\beta}}$ is unbiased for $\boldsymbol{\beta}$

Fitted values are given by

$$\hat{\boldsymbol{Y}} \;=\; X\hat{\boldsymbol{\beta}} \;=\; X\left(X^T X\right)^{-1} X^T \boldsymbol{Y} \;=\; H\boldsymbol{Y}$$

where $H = X\left(X^T X\right)^{-1} X^T$ is called the **hat matrix**.

The Residual (or Error) Sum of Squares is

$$\begin{aligned} SS_{RES} &= \left(\boldsymbol{Y} - X\hat{\boldsymbol{\beta}}\right)^T \left(\boldsymbol{Y} - X\hat{\boldsymbol{\beta}}\right) \\ &= \boldsymbol{Y}^T\boldsymbol{Y} - 2\hat{\boldsymbol{\beta}}^T X^T\boldsymbol{Y} + \hat{\boldsymbol{\beta}}^T X^T X\hat{\boldsymbol{\beta}} \\ &= \boldsymbol{Y}^T\boldsymbol{Y} - 2\hat{\boldsymbol{\beta}}^T X^T\boldsymbol{Y} + \hat{\boldsymbol{\beta}}^T X^T X\left(X^T X\right)^{-1} X^T\boldsymbol{Y} \\ &= \boldsymbol{Y}^T\boldsymbol{Y} - \hat{\boldsymbol{\beta}}^T X^T\boldsymbol{Y} \end{aligned}$$

We can estimate σ^2 based on this:

$$\boxed{\widehat{\sigma^2} = \frac{1}{n-(k+1)}\, SS_{RES}}$$

(n observations, k explanatory variables.)

Note that $\widehat{\sigma^2}$ is unbiased (unlike the MLE).